

# LC26G&LC76G&LC86G Series

## Low Power Mode Application Note

**GNSS Module Series**

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# About the Document

## Document Information

|                 |                                                          |
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## Revision History

| Version | Date       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| -       | 2023-02-17 | Creation of the document                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 1.0     | 2023-04-27 | First official release                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1.1     | 2024-05-13 | <ol style="list-style-type: none"> <li>Added applicable module LC86G (PA) (<a href="#">Chapter 1</a>).</li> <li>Added Balloon and Stationary modes (<a href="#">Table 2</a> and <a href="#">Table 3</a>).</li> <li>Updated constellation options of ALP, GLP and FLP modes (<a href="#">Table 3</a>).</li> <li>Updated testing procedure of the Backup Mode (<a href="#">Chapter 2.1.2</a>).</li> <li>Updated the mechanism and testing procedure of the ALP mode (<a href="#">Chapter 2.3</a>).</li> <li>Updated the testing procedure of the GLP mode (<a href="#">Chapter 2.4.2</a>).</li> <li>Updated the testing procedure of the FLP mode (<a href="#">Chapter 2.5.2</a>).</li> <li>Added EPH and ALM related messages (<a href="#">Chapter 3</a>).</li> </ol> |

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# 1 Introduction

The LC26G (AB), LC76G series and LC86G series modules support different low power modes with enhanced power consumption and outstanding positioning performances. This document describes the operation mechanisms and related testing procedures of Backup, Periodic Power Saving, Adaptive Low Power (ALP), GPS Low Power (GLP), and Fitness Low Power (FLP) modes, as well as EPH and ALM related messages.

In Backup mode, the module is in an inactive state where most functions are disabled, and the power consumption mainly comes from the RTC circuit.

Periodic Power Saving mode consists of running (active) and sleeping stages, whose time span can be adjusted to reduce power consumption.

FLP mode extends GLP mode by adding more constellation combinations, while ALP mode is a subset of FLP mode that differs in navigation mode. The differences between GLP, FLP and ALP modes are detailed in [Table 3: Low Power Modes and Corresponding Configurations](#).

The commands related to low power modes are listed below. Before implementing any of the commands below, refer to [document \[1\] protocol specification](#).

**Table 1: Commands Related to Low Power Modes**

| PAIR Command | Description                                                                                                                      |
|--------------|----------------------------------------------------------------------------------------------------------------------------------|
| PAIR050      | Sets the position fix interval.                                                                                                  |
| PAIR051      | Gets the position fix interval.                                                                                                  |
| PAIR066      | Sets the GNSS search mode.                                                                                                       |
| PAIR067      | Gets the GNSS search mode.                                                                                                       |
| PAIR080      | Sets the navigation mode.                                                                                                        |
| PAIR081      | Gets the navigation mode.                                                                                                        |
| PAIR650      | Powers off the GNSS system, except the clock. This command sets the CPU to Backup mode, in which it cannot receive any commands. |
| PAIR680      | Enables/disables GLP mode.                                                                                                       |



| PAIR Command | Description                                         |
|--------------|-----------------------------------------------------|
| PAIR681      | Gets GLP mode setting.                              |
| PAIR690      | Sets the Periodic Power Saving mode configurations. |
| PAIR691      | Gets the Periodic Power Saving mode configurations. |
| PAIR730      | Enables/disables FLP mode.                          |
| PAIR731      | Gets FLP mode setting.                              |
| PAIR732      | Enables/disables ALP mode.                          |
| PAIR733      | Gets ALP mode setting.                              |

**Table 2: Navigation Modes**

| Navigation Mode | Description                                                                                                                                          |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Normal mode     | Used for general purposes.                                                                                                                           |
| Fitness mode    | Used for running and walking purposes, making low-speed movement (< 5 m/s) more impactful on position calculation.                                   |
| Swimming mode   | Used for swimming purposes to smooth the trajectory and improve the accuracy of distance calculation.                                                |
| Drone mode      | Used for drone applications with equivalent dynamic range and vertical acceleration at different flight phases (for example, hovering and cruising). |
| Balloon mode    | Used for high-altitude balloon scenario where the vertical movement has a greater impact on the position calculation.                                |
| Stationary mode | Used for stationary applications where a zero dynamic is assumed.                                                                                    |

The corresponding configurations of the low power modes supported by LC26G (AB), LC76G series and LC86G series are as follows:

**Table 3: Low Power Modes and Corresponding Configurations**

| Type                       | Navigation Mode   | Constellation Option <sup>1)</sup>                                                                                                | Fix Update Rate <sup>2)</sup> | Supported Module         |
|----------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------------------|
| Backup mode                |                   | <ul style="list-style-type: none"> <li>● GPS only</li> <li>● GPS + QZSS</li> <li>● GPS + GLONASS</li> </ul>                       |                               |                          |
|                            | ● Normal mode     | ● GPS + GLONASS + QZSS                                                                                                            |                               |                          |
|                            | ● Fitness mode    | ● GPS + Galileo                                                                                                                   | ● 1 Hz                        | ● LC26G (AB)             |
|                            | ● Swimming mode   | ● GPS + Galileo + QZSS                                                                                                            | ● 2 Hz                        | ● LC76G (AB, PA, PB)     |
|                            | ● Drone mode      | ● GPS + BDS                                                                                                                       | ● 5 Hz                        | ● LC86G (AA, AB, LA, PA) |
|                            | ● Balloon mode    | ● GPS + BDS + QZSS                                                                                                                | ● 10 Hz                       |                          |
|                            | ● Stationary mode | <ul style="list-style-type: none"> <li>● GPS + GLONASS + Galileo + BDS</li> <li>● GPS + GLONASS + Galileo + BDS + QZSS</li> </ul> |                               |                          |
| Periodic Power Saving mode |                   | <ul style="list-style-type: none"> <li>● GPS only</li> <li>● GPS + QZSS</li> <li>● GPS + GLONASS</li> </ul>                       |                               |                          |
|                            | ● Normal mode     | ● GPS + GLONASS + QZSS                                                                                                            |                               |                          |
|                            | ● Fitness mode    | ● GPS + Galileo                                                                                                                   | ● 1 Hz                        | ● LC26G (AB)             |
|                            | ● Swimming mode   | ● GPS + Galileo + QZSS                                                                                                            | ● 2 Hz                        | ● LC76G (AB, PA, PB)     |
|                            | ● Drone mode      | ● GPS + BDS                                                                                                                       | ● 5 Hz                        | ● LC86G (AA, AB, LA, PA) |
|                            | ● Balloon mode    | ● GPS + BDS + QZSS                                                                                                                | ● 10 Hz                       |                          |
|                            | ● Stationary mode | <ul style="list-style-type: none"> <li>● GPS + GLONASS + Galileo + BDS</li> <li>● GPS + GLONASS + Galileo + BDS + QZSS</li> </ul> |                               |                          |

| Type                                  | Navigation Mode | Constellation Option <sup>1)</sup>                                                                                                                                                                                                                                                                                                             | Fix Update Rate <sup>2)</sup> | Supported Module                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ALP mode<br>(ALP mode 1 & ALP mode 2) | ● Normal mode   | <ul style="list-style-type: none"> <li>● GPS only</li> <li>● GPS + QZSS</li> <li>● GPS + GLONASS</li> <li>● GPS + GLONASS + QZSS</li> <li>● GPS + Galileo</li> <li>● GPS + Galileo + QZSS</li> <li>● GPS + BDS</li> <li>● GPS + BDS + QZSS</li> <li>● GPS + GLONASS + Galileo + BDS</li> <li>● GPS + GLONASS + Galileo + BDS + QZSS</li> </ul> | ● 1 Hz                        | <b>ALP mode 1:</b> <ul style="list-style-type: none"> <li>● LC26G (AB) with LC26GABNR12A01S or higher versions</li> <li>● LC76G (AB) with LC76GABNR12A01S, LC76G (PA) with LC76GPANR02A02S and LC76G (PB) with LC76GPBNR02A02S or higher versions</li> <li>● LC86G (AA, AB, PA) and LC86G (LA) with LC86GLANR02A02S or higher versions</li> </ul> <b>ALP mode 2:</b> <ul style="list-style-type: none"> <li>● LC26G (AB) with LC26GABNR12A02S or higher versions</li> <li>● LC76G (AB) with LC76GABNR12A02S, LC76G (PA) with LC76GPANR02A03S and LC76G (PB) with LC76GPBNR12A02S or higher versions</li> <li>● LC86G (PA) and LC86G (AA) with LC86GAANR12A03S and LC86G (LA) with LC86GLANR12A02S or higher versions</li> </ul> |
| GLP mode                              | ● Fitness mode  | <ul style="list-style-type: none"> <li>● GPS only</li> <li>● GPS + QZSS</li> </ul>                                                                                                                                                                                                                                                             | ● 1 Hz                        | <ul style="list-style-type: none"> <li>● LC26G (AB)</li> <li>● LC76G (AB, PA, PB)</li> <li>● LC86G (AA, AB, LA, PA)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| FLP mode                              | ● Fitness mode  | <ul style="list-style-type: none"> <li>● GPS + GLONASS</li> <li>● GPS + GLONASS + QZSS</li> <li>● GPS + BDS</li> <li>● GPS + BDS + QZSS</li> </ul>                                                                                                                                                                                             | ● 1 Hz                        | <ul style="list-style-type: none"> <li>● LC26G (AB)</li> <li>● LC76G (AB, PA, PB)</li> <li>● LC86G (AA, AB, LA, PA)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| Type | Navigation Mode | Constellation Option <sup>1)</sup>                                                                                                | Fix Update Rate <sup>2)</sup> | Supported Module |
|------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------|------------------|
|      |                 | <ul style="list-style-type: none"> <li>● GPS + GLONASS + Galileo + BDS</li> <li>● GPS + GLONASS + Galileo + BDS + QZSS</li> </ul> |                               |                  |

#### NOTE

1. The navigation mode, GNSS satellite configuration and fix update rate are the prerequisites for the operation of the low power modes. If the module is not in tracking mode that matches one of the available navigation profiles, it cannot support low power modes.
2. <sup>1)</sup> GLONASS is not supported on LC86G (AA) and BDS is not supported on LC86G (AB). For details of their constellation configurations, see [document \[1\] protocol specification](#).
3. <sup>2)</sup> For LC76G (PA, PB) and LC86G (PA) modules, the fix update rate can only be 1 Hz.

## 2 Low Power Modes

To extend battery operation time and meet customer requirements, the module supports several low power modes.

**Table 4: System Resources Utilized in Low Power Modes**

| System Resource   | Backup | Periodic Power Saving (Running Stage) | Periodic Power Saving (Sleeping Stage) | ALP | GLP | FLP |
|-------------------|--------|---------------------------------------|----------------------------------------|-----|-----|-----|
| RTC Macro         | ●      | ●                                     | ●                                      | ●   | ●   | ●   |
| RTC SRAM          | ●      | ●                                     | ●                                      | ●   | ●   | ●   |
| Internal Circuits | -      | ●                                     | ●                                      | ●   | ●   | ●   |
| CPU core          | -      | ●                                     | ●                                      | ●   | ●   | ●   |
| SRAM              | -      | ●                                     | ●                                      | ●   | ●   | ●   |
| TCXO              | -      | ●                                     | ●                                      | ●   | ●   | ●   |
| GPIO              | -      | ●                                     | ●                                      | ●   | ●   | ●   |
| RF                | -      | ●                                     | -                                      | ●   | ●   | ●   |

### 2.1. Backup Mode

#### 2.1.1. Operation Mechanism

Backup mode is used to improve power efficiency. It shuts down most system components, such as RF controller, processing engine DSP, CPU core and so on. As a result, while the module is in Backup mode, the module's communication port is unresponsive to any incoming commands and no data is transmitted over it. Before it enters the Backup mode, the module saves navigation data, including ephemeris, almanac, location and time to RTC SRAM for the next navigation cycle to shorten the TTFF. The CPU core can be awoken by either setting the timer or powering on the VCC pin. For details, refer to [document \[1\] protocol specification](#) and [documents \[2\], \[3\], and \[4\] hardware designs](#).

Once the CPU core exits the Backup mode, the navigation data will be loaded from RTC SRAM. If all the

data is valid, a shorter TTFF can be achieved.

### 2.1.2. Testing Procedure

To Enter the Backup mode:

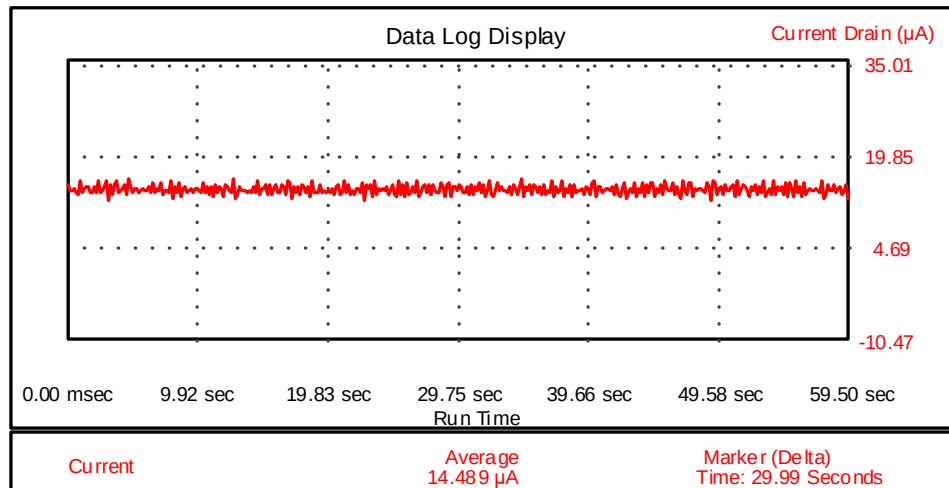
1. Send **\$PAIR650** command;
2. Disconnect VCC power supply and keep the V\_BCKP pin powered.

To exit the Backup mode:

- If **<Second>** is set to 0, restore VCC. (For LC76G (PB), it needs to be in Backup mode for at least 10 s.)
- If **<Second>** is not 0 and the time has not expired, restore the VCC power supply first and then pull the RESET\_N pin low for at least 100 ms.
- If **<Second>** is not 0 and the time expires, restore the VCC power.

**Table 5: Testing Environment in Backup Mode**

| Item               | Description                                                    |
|--------------------|----------------------------------------------------------------|
| Module             | LC76G (PA)                                                     |
| Voltage            | 3.3 V                                                          |
| RF Signal Strength | -130 dBm                                                       |
| Equipment          | Keysight 66319B                                                |
| Command            | <b>\$PAIR650,0*25</b>                                          |
| Execution Time     | After the module gets a fix.                                   |
| Axis               | Vertical axis: current drain (μA)<br>Horizontal axis: run time |



**Figure 1: Measured Power Consumption in Backup Mode**

## 2.2. Periodic Power Saving Mode

### 2.2.1. Operation Mechanism

In the Periodic Power Saving mode, power consumption can be reduced by turning off the DSP and periodically putting the CPU core to sleep. Average consumption is conditioned by the duration of running and sleeping times, which can be adjusted as needed.

- During the running time, the DSP and CPU core are active and perform their respective tasks, while the GNSS module generates the position-related information.
- During the sleeping time, the DSP is paused while the CPU core is in the sleeping stage.

The time span of the running (active) and sleeping stages can be dynamically changed or strictly limited, i.e., Smart Periodic Mode and Strict Periodic Mode. In Smart Periodic mode, the module observes the current satellite signal status, satellite health status, collected ephemeris status, etc., to determine the time needed to collect ephemeris data. Therefore, when **\$PAIR690** is implemented, the system will automatically adjust the extension time to collect the required ephemeris data, increasing the likelihood that the position will be available immediately after the module wakes up from sleep. To shorten TTFF after waking up, it is recommended to use Smart Periodic mode to dynamically change the time span of the running stage to obtain more navigation information, such as pseudorange, ephemeris, and almanac. In Strict Periodic mode the duration of every cycle is fixed because the time span of the running stage is limited to a specific value. This fixed time span means that the module may not be able to complete satellite data update, such as ephemeris, within a single cycle, could potentially affect the TTFF.

Periodic pattern schematic diagram with different colors representing specific stages:

- Blue: module is in running (active) stage and the GNSS signal is good.
- Red: module is in sleeping stage and the GNSS signal is good.

- Yellow: module is in running (active) stage and there is no GNSS signal.
- Green: module is in sleeping stage and there is no GNSS signal.
- Brown: the extension time of running (active) stage.

Module operation in Smart Periodic mode is illustrated in the following figure:

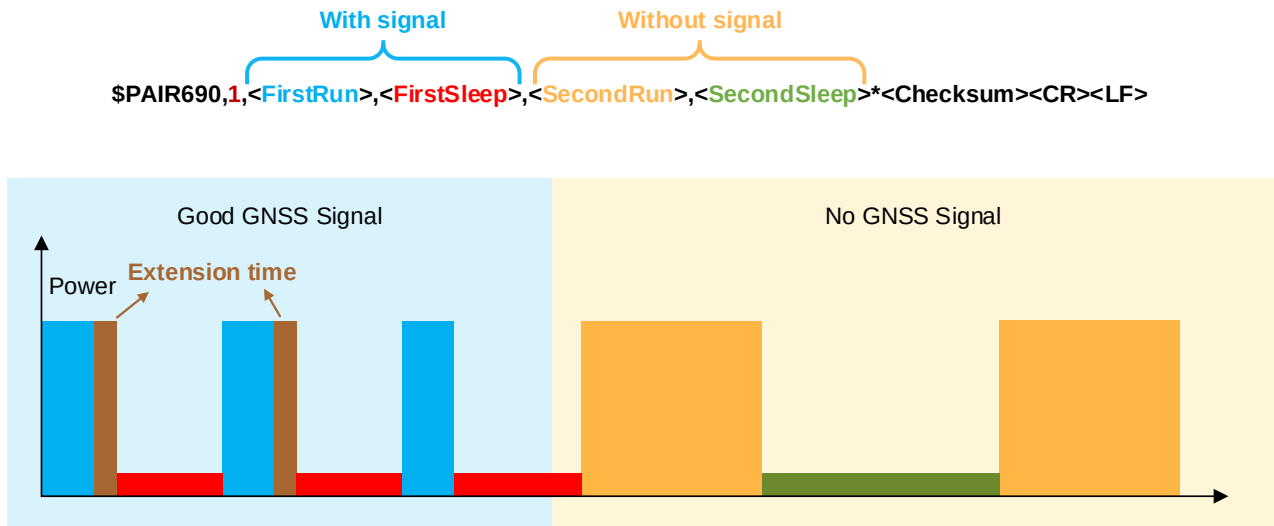


Figure 2: Module Operation in Smart Periodic Mode

Module operation in Strict Periodic mode is illustrated in the following figure:

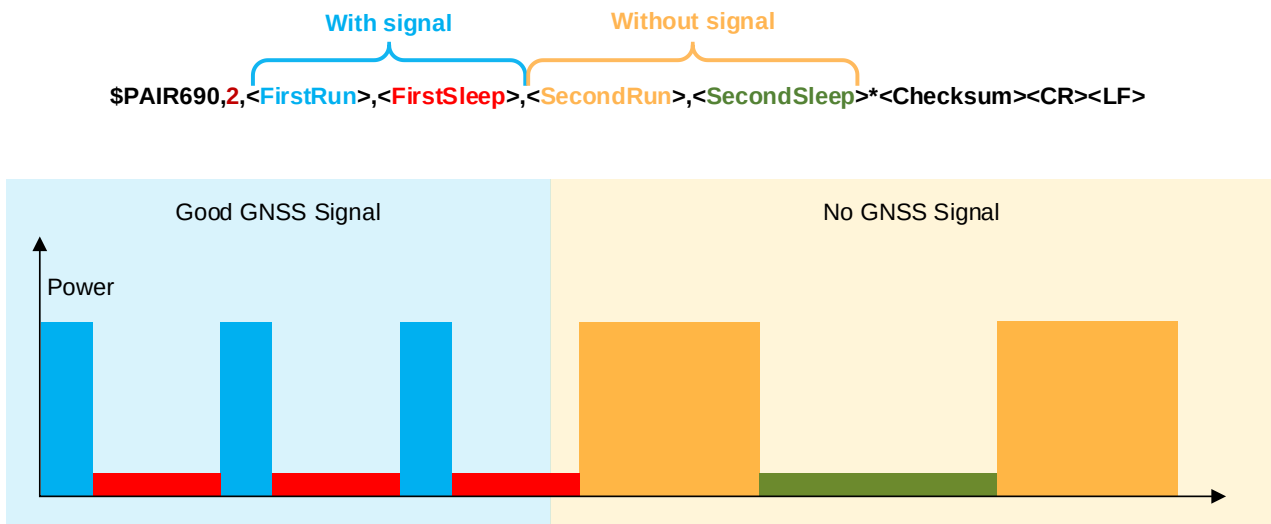


Figure 3: Module Operation in Strict Periodic Mode



### 2.2.2. Testing Procedure

To enter the Periodic Power Saving mode:

1. Power on the module.
2. Send **\$PAIR690** to enter the Periodic Power Saving mode.

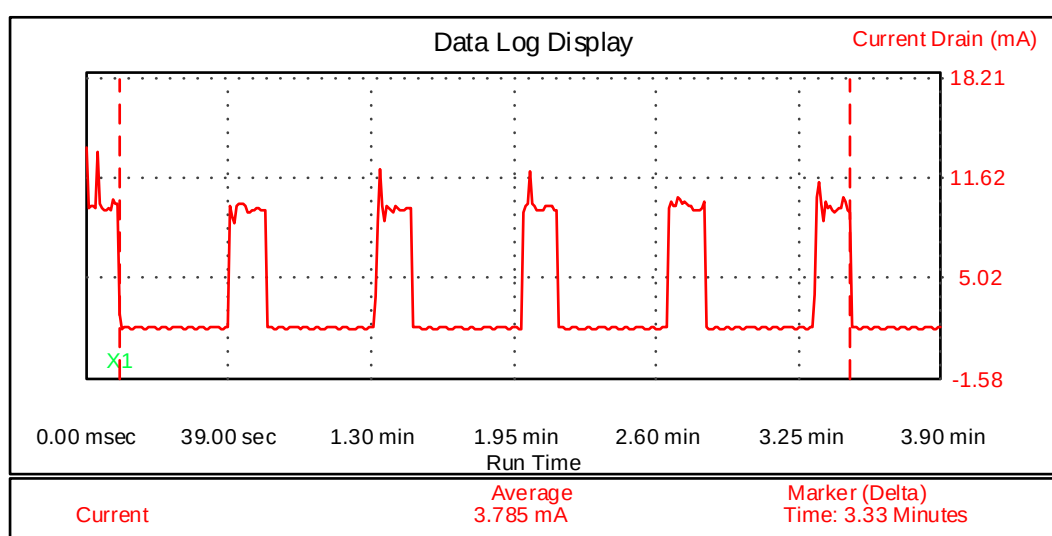
The module returns **\$PAIR001,690,0\*34** when it enters the Periodic Power Saving mode.

To exit the Periodic Power Saving mode:

Send **\$PAIR690,0\*29** to exit the Periodic Power Saving mode.

**Table 6: Testing Environment in Periodic Power Saving Mode**

| Item               | Description                                                    |
|--------------------|----------------------------------------------------------------|
| Module             | LC76G (PA)                                                     |
| Voltage            | 3.3 V                                                          |
| RF Signal Strength | -130 dBm                                                       |
| Equipment          | Keysight 66319B                                                |
| Command            | <b>\$PAIR690,2,10,30,30,40*2E</b>                              |
| Execution Time     | After the module gets a fix.                                   |
| Axis               | Vertical axis: current drain (mA)<br>Horizontal axis: run time |

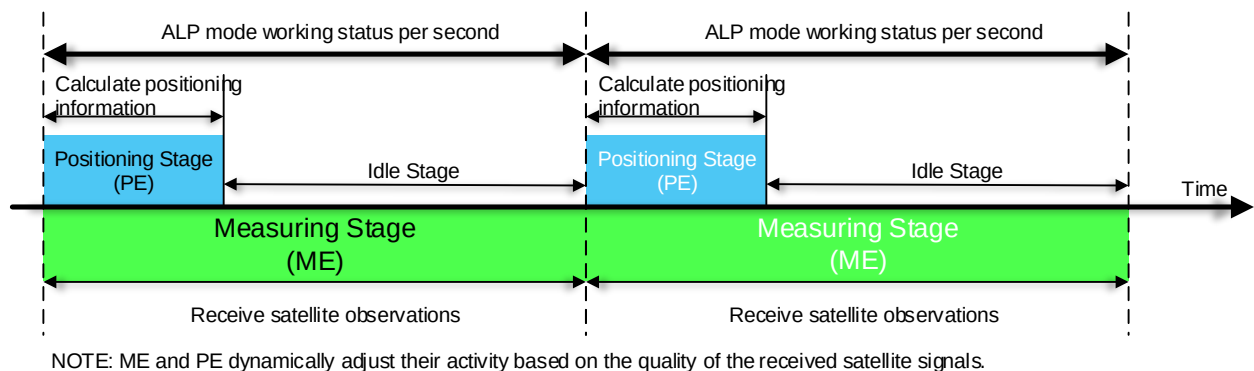


**Figure 4: Measured Power Consumption in Periodic Power Saving Mode**

## 2.3. ALP Mode (ALP Mode 1 & ALP Mode 2)

### 2.3.1. Operation Mechanism

In the ALP mode, power is saved by using a method similar to duty cycle, but the position quality is degraded compared to that in tracking mode (in the Continuous mode). The receiver outputs a positioning solution at every epoch, but with lower precision. The ALP mode is suitable for scenarios where long-term module operation and power efficiency are critical. However, it is not suitable for scenarios where short-term operation and high positioning accuracy are prioritized.



**Figure 5: Module Operation in ALP Mode**

As illustrated above, the ALP mode consists of three stages:

- Measuring Stage: ME receives satellite observations.
- Positioning Stage: PE calculates positioning information.
- Idle Stage: Module is in idle stage.

Upon initial power-on and entering the ALP mode, the module does not immediately reduce power consumption. Instead, it uses the ME and PE to actively search for all satellites in view, ensuring the shortest possible TTFF and the highest sensitivity for satellite signal reception. As the module downloads enough ephemeris and almanac data, it gradually reduces power consumption. Once both the ephemeris and almanac data are fully downloaded, the power consumption of the module stabilizes at its optimal level for the ALP mode. The detailed process is as follows:

1. To achieve a good positioning performance after startup, the module continuously searches for GNSS satellite signals, and its ME and PE consume most of the power.
2. As the module accumulates enough GNSS satellite data, it gradually reduces the usage of ME and PE, resulting in a gradual reduction in power consumption.
3. After downloading the complete almanac (which takes about 12.5 minutes) the power consumption of the module stabilizes. The ME and PE then dynamically adjust their activity based on the quality of the received satellite signals to maintain position accuracy while optimizing power usage.

In the ALP mode, the module searches for satellites and adjusts the search time according to the actual conditions to achieve better positioning results while minimizing power consumption. This means that the satellite search time can be extended when satellite availability and signal quality are weak and the satellite search time can be shortened when satellite availability and signal quality are good.

### 2.3.1.1. Comparison Between ALP Mode and Continuous Mode

ALP mode offers two options that prioritize either power efficiency or positioning performance:

- ALP Mode 1 (power mode);
- ALP Mode 2 (performance mode).

The following table compares these ALP modes with the Continuous mode.

**Table 7: Module Performance in Continuous Mode, ALP Mode 1 and ALP Mode 2**

| Feature              | Continuous Mode | ALP Mode 1          | ALP Mode 2                                        |
|----------------------|-----------------|---------------------|---------------------------------------------------|
| Power Consumption    | Highest         | Lowest              | Moderate (between Continuous mode and ALP Mode 1) |
| Positioning Accuracy | Best            | Diminished          | Moderate (between Continuous mode and ALP Mode 1) |
| Satellite Search     | Not limitations | Severe limitations  | Slightly limited                                  |
| RF Circuit Activity  | Unrestricted    | Dynamically adjusts | Unrestricted                                      |

### 2.3.1.2. Shorten Time to Achieve Low Power Consumption in ALP Mode

The module reduces power consumption in ALP mode when the following conditions are met:

- A valid position is obtained.
- The entire almanac has been downloaded (about 12.5 minutes).
- The ephemeris for each satellite in view is valid.

To shorten the time to achieve low power consumption in ALP mode, consider these options:

- **Save ephemeris and almanac data to flash.** Send **\$PAIR511\*3F** to save the ephemeris and almanac data to flash memory after each satellite search or before the module is turned off, to avoid data loss due to expired ephemeris data or a power failure.
- **Use the Backup mode instead of powering down the module.** The module saves satellite data to RTC RAM in the Backup mode. By waking the module from the Backup mode and then entering the

ALP mode (provided the satellite data is still valid), the time required to achieve low power consumption in the ALP mode can be reduced.

- **Use external ephemeris and almanac data.** If another device has valid ephemeris and almanac data for the satellites in view, send the data to the module to significantly reduce the time for achieving low power consumption in ALP mode. For relevant commands, refer to [Chapter 3 EPH and ALM Related Messages](#).
- **Inject AGNSS EPO files.** Injecting EPO files into the module using the AGNSS service can also reduce the time for achieving low power consumption in ALP mode. This is an auxiliary method that is effective to some degree, but may not be as efficient as the other options. For details on EPO file injection, refer to [document \[5\] AGNSS application note](#).

### 2.3.2. Testing Procedure

Please refer to [Table 3: Low Power Modes and Corresponding Configurations](#) before performing the following test.

To enter the ALP mode (when the module is in default configurations):

1. Power on the module.
2. Send **\$PAIR080** to set navigation mode to the Normal mode.
3. Send **\$PAIR732** to enter the ALP mode.

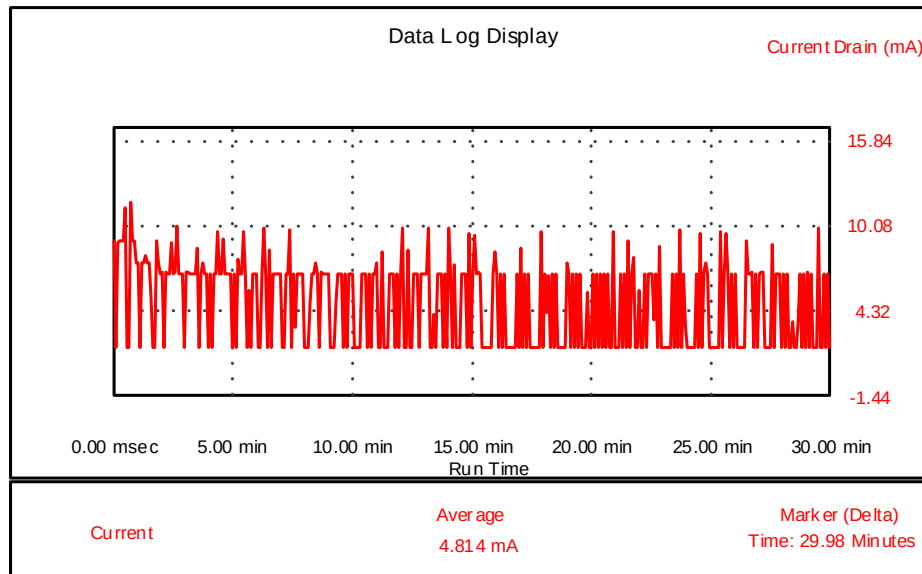
The module returns **\$PAIR001,732,0\*3D** when it enters the ALP mode.

To exit the ALP mode:

Send **\$PAIR732,0\*20** to exit the ALP mode.

**Table 8: Testing Environment in ALP Mode 1 – Entering ALP Mode 1 After Power On**

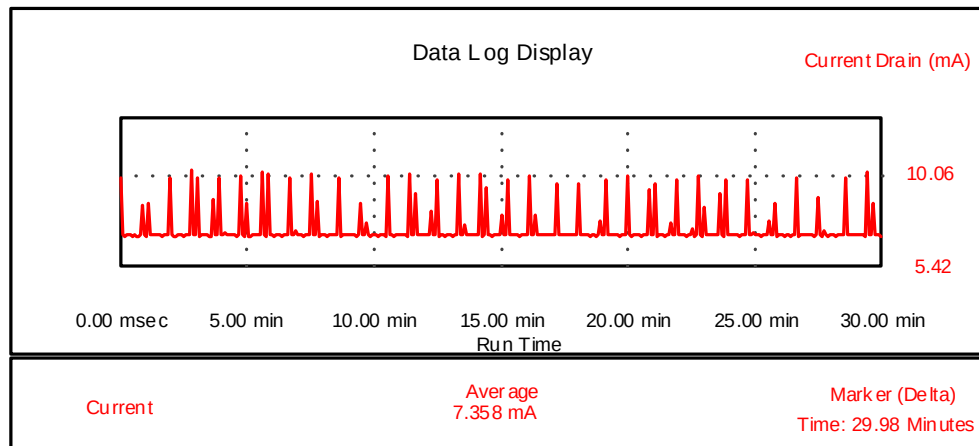
| Item               | Description                                                    |
|--------------------|----------------------------------------------------------------|
| Module             | LC76G (PA)                                                     |
| Voltage            | 3.3 V                                                          |
| RF Signal Strength | -130 dBm                                                       |
| Equipment          | Keysight 66319B                                                |
| Command            | <b>\$PAIR732,1*21</b>                                          |
| Execution Time     | Immediately after the module is powered on.                    |
| Axis               | Vertical axis: current drain (mA)<br>Horizontal axis: run time |



**Figure 6: Power Consumption Measured in ALP Mode 1 After Power On**

**Table 9: Testing Environment in ALP Mode 2 – Entering ALP Mode 2 After Power On**

| Item               | Description                                                    |
|--------------------|----------------------------------------------------------------|
| Module             | LC76G (PA)                                                     |
| Voltage            | 3.3 V                                                          |
| RF Signal Strength | -130 dBm                                                       |
| Equipment          | Keysight 66319B                                                |
| Command            | <b>\$PAIR732,2*22</b>                                          |
| Execution Time     | Immediately after the module is powered on.                    |
| Axis               | Vertical axis: current drain (mA)<br>Horizontal axis: run time |

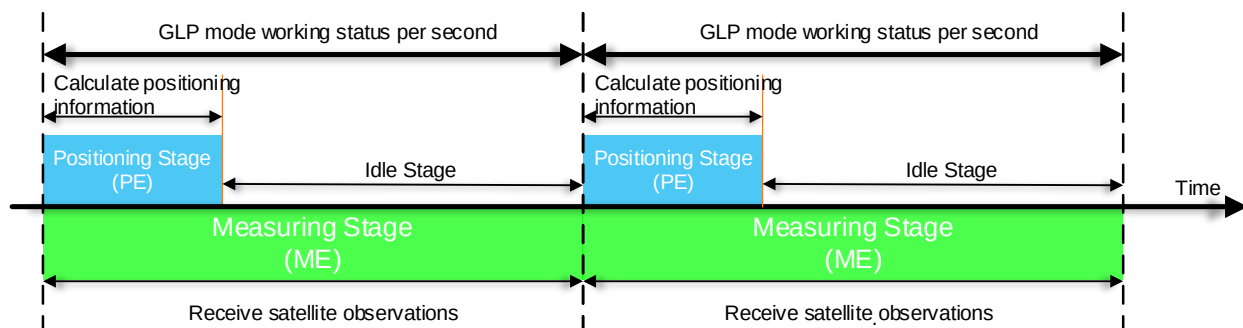


**Figure 7: Power Consumption Measured in ALP Mode 2 After Power On**

## 2.4. GLP Mode

### 2.4.1. Operation Mechanism

In GLP mode, power is saved by using a method similar to duty cycle, but the position quality is degraded compared to that in tracking mode (in the Continuous mode).



### Figure 8: Module Operation in GLP Mode

As illustrated above, the GLP mode consists of three stages:

- Measuring Stage: ME receives satellite observations.
- Positioning Stage: PE calculates positioning information.
- Idle Stage: the module is in idle stage.

When the module enters GLP mode, it will further limit the search for GPS satellites, reducing the time required to acquire satellite signals and parse information, while maintaining low power consumption.

## 2.4.2. Testing Procedure

Please refer to [Table 3: Low Power Modes and Corresponding Configurations](#) before performing the following test.

To enter the GLP mode (when the module is in default configurations):

1. Power on the module.
2. Send **\$PAIR066** to set the GNSS search mode to GPS only or GPS + QZSS.
3. Send **\$PAIR080** to set navigation mode to the Fitness mode.
4. Send **\$PAIR680** to enter the GLP mode.

The module returns **\$PAIR001,680,0\*35** when it enters the GLP mode.

To exit the GLP mode:

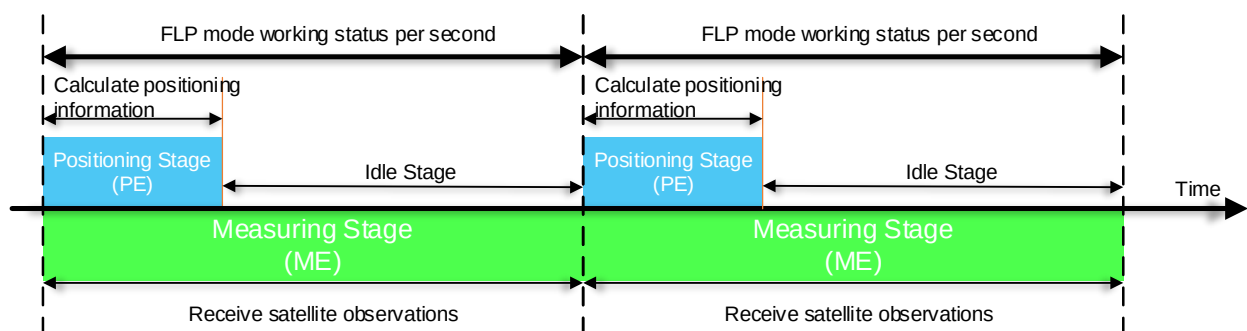
Send the **\$PAIR680,0\*28** to exit the GLP mode.

The module's power consumption in the GLP mode is similar to that in the ALP Mode 1. For the description of power consumption in the ALP mode 1, please refer to [Chapter 2.3.2 Testing Procedure](#).

## 2.5. FLP Mode

### 2.5.1. Operation Mechanism

In FLP mode, power is saved by using a method similar to duty cycle, but the position quality is degraded compared to that in tracking mode (in the Continuous mode).



**Figure 9: Module Operation in FLP Mode**

As shown above, the FLP mode consists of three stages:

- Measuring Stage: ME receives satellite observations.
- Positioning Stage: PE calculates positioning information.
- Idle Stage: the module is in idle stage.

Due to the satellite search limitations in FLP mode, the positioning accuracy is lower compared to that in the Continuous mode. However, because the module limits the number of satellites it searches, the time for satellite search and calculation is reduced, resulting in lower power consumption in this mode.

### **2.5.2. Testing Procedure**

Please refer to [Table 3: Low Power Modes and Corresponding Configurations](#) before performing the following test.

To enter the FLP mode (when the module is in default configurations):

1. Power on the module.
2. Send **\$PAIR080** to set navigation mode to the Fitness mode.
3. Send **\$PAIR730** to enter the FLP mode.

The module returns **\$PAIR001,730,0\*3F** when it enters the FLP mode.

To exit the FLP mode:

Send **\$PAIR730,0\*22** to exit the FLP mode.

The module's power consumption in FLP mode is similar to that in ALP Mode 1. For the description of power consumption in the ALP mode 1, please refer to [Chapter 2.3.2 Testing Procedure](#).



# 3 EPH and ALM Related Messages

This chapter explains the EPH and ALM related messages supported by the module.

## NOTE

For details about \$PAIR001 message, see [document \[1\] protocol specifications](#).

## 3.1. EPH Related Messages

### 3.1.1. PAIR530: PAIR\_EPH\_GET\_STATUS

Queries the satellites whose valid EPH data will be available for a specified future time period.

#### Type:

Command

#### Synopsis:

```
$PAIR530,<Constellation>,<Time_Interval>*<Checksum><CR><LF>
```

#### Parameter:

| Field           | Format  | Unit   | Description                                                                     |
|-----------------|---------|--------|---------------------------------------------------------------------------------|
| <Constellation> | Numeric | -      | GNSS System ID.<br>0 = GPS<br>1 = GLONASS<br>2 = Galileo<br>3 = BDS<br>4 = QZSS |
| <Time_Interval> | Numeric | Second | Specified future time period. Range: 1–7200 (2 hours).                          |

#### Result:

Returns \$PAIR001 message and the query result.

### Query result message format:

```
$PAIR530,<Constellation>,<L1_SV>*<Checksum><CR><LF>
```

### Parameters included in the result:

| Field           | Format      | Unit | Description                                                                                                                                                                                                                                                                                                                                                         |
|-----------------|-------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Constellation> | Numeric     | -    | GNSS System ID.<br>0 = GPS<br>1 = GLONASS<br>2 = Galileo<br>3 = BDS<br>4 = QZSS                                                                                                                                                                                                                                                                                     |
| <L1_SV>         | Hexadecimal | -    | Valid EPH SV ID. Field Lengths for different constellations are as follows.<br>GPS: 8 bytes<br>GLONASS: 6 bytes<br>Galileo: 16 bytes<br>BDS: 16 bytes<br>QZSS: 8 bytes<br>Note that each bit represents a satellite, and SV ID counting starts from the least significant bit. Take GPS as an example. If Bit 0 is 1, the EPH data of GPS satellite 1 is available. |

### Example:

```
$PAIR530,0,10*0D
```

```
$PAIR001,530,1*3C
```

```
$PAIR001,530,0*3D
```

```
$PAIR530,0,DA008004*05
```

//The binary format of DA008004 is: 1101 1010 0000 0000 1000 0000 0000 0100. The corresponding satellite numbers are 3, 16, 26, 28, 29, 31, and 32.

### 3.1.2. PAIR531: PAIR\_EPH\_SET\_DATA

Sends EPH data to GNSS module.

### Type:

Input

### Synopsis:

```
$PAIR531,<Constellation>,<Signal_ID>,<SV_ID>,<EPH_Data>*<Checksum><CR><LF>
```

### Parameter:

| Field           | Format      | Unit | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------|-------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Constellation> | Numeric     | -    | GNSS System ID.<br>0 = GPS<br>1 = GLONASS<br>2 = Galileo<br>3 = BDS<br>4 = QZSS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <Signal_ID>     | Numeric     | -    | Signal ID. Always "0".<br>0 = L1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <SV_ID>         | Numeric     | -    | SV ID. Ranges for different constellations are as below.<br>GPS: 1–32<br>GLONASS: 1–24<br>Galileo: 1–36<br>BDS: 1–63 (GEO: 1–5, 59–63; MEO: 6–58)<br>QZSS: 1–7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <EPH_Data>      | Hexadecimal | -    | <p>EPH data. The format of EPH data for different constellations is as follows.</p> <p><b>GPS and QZSS:</b> Word[0]–Word[23]. The 24-word EPH data cover words 3–10 from subframes 1–3 in the GPS or QZSS Navigation Message. The raw 30-bit data of each word have been logically shifted 6 bits to the right to remove the 6 parity bits leaving the 24-bit data. For details on data format, see <a href="#">Figure 10: EPH Data Format – GPS and QZSS</a>.</p> <p><b>GLONASS:</b> Word[0]–Word[15]. The first 15-word EPH data cover strings 1–5 in the GLONASS Navigation Message. The Last word, Word[15], indicates the frequency channel (range: 1–14). For details on data format, see <a href="#">Figure 11: EPH Data Format – GLONASS</a>.</p> <p><b>Galileo:</b> Word[0]–Word[18]. The 19-word EPH data cover 16 words from the subframes 1–4 in the Galileo Navigation Message and 3 words from subframe 5 with BGD, signal health status, data validity, and GST. Note that only Word[0]–Word[15] need to be input when sending ephemeris, while Word[0]–Word[18] will be provided</p> |

| Field | Format | Unit | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------|--------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |        |      | <p>when getting ephemeris.</p> <p>The Page Type (6-bit) and IOD<sub>nav</sub> (10-bit) have been removed and the rest bits of each word are shifted to the right. For details on data format, see <a href="#">Figure 12: EPH Data Format – Galileo</a>.</p> <p><b>BDS GEO:</b> Word[0]–Word[29]. The 30-word EPH data cover pages 1–10 from subframe 1 in the BDS GEO D2 Navigation Message.</p> <p>The raw 30-bit data of each word have been logically shifted 8 bits to the right to remove the 8 parity bits. For details on data format, see <a href="#">Figure 13: EPH Data Format – BDS GEO</a>.</p> <p><b>BDS MEO:</b> Word[0]–Word[20]. The 21-word EPH data cover subframes 1–3 in the BDS MEO D1 Navigation Message.</p> <p>The raw 30-bit data of each word have been logically shifted 8 bits to the right to remove the 8 parity bits. For details on data format, see <a href="#">Figure 14: EPH Data Format – BDS MEO</a>.</p> |

**Result:**

Returns **\$PAIR001** message and the query result.

**Query result message format:**

```
$PAIR531,<Status>*<Checksum><CR><LF>
```

**Parameter included in the result:**

| Field    | Format  | Unit | Description                  |
|----------|---------|------|------------------------------|
| <Status> | Numeric | -    | 0 = Failed<br>1 = Successful |

**Example:**

```
//GPS
$PAIR531,0,0,05,369000,138AF9,101EA3,3338BA,78CBE8,1A60AE,00FFF3,EECD7B,1AF784,2C3D7
E,190507,F8C903,141524,1631A1,0D7E3F,60AE7C,002C7A,467A1D,FFF527,62DB0B,16562F,489AE
F,FFAD58,1A0AD4*68
$PAIR001,531,1*3D
$PAIR001,531,0*3C
$PAIR531,1*20
```

```
//GLONASS
$PAIR531,1,0,03,079AACFF,8305482D,0000009F,17A0921B,8580D948,00000013,80168FB6,A29AB6
66,00000075,8539BE60,00026277,0000000D,9DD00000,00238000,00000062,0000000D*60
$PAIR001,531,1*3D
$PAIR001,531,0*3C
$PAIR531,1*20

//Galileo
$PAIR531,2,0,10,49E8DAC4,D9680009,DDFEA813,00007E0F,A98274AD,28BD1AFA,D8AE6A95,000
00621,FFC67C1A,B401BA14,87130802,00004A6B,280017FF,FD27AFEC,652CDFFF,00003301,16401
00C,04FB45AA,0000000B*4C
$PAIR001,531,1*3D
$PAIR001,531,0*3C
$PAIR531,1*20

//BDS GEO
$PAIR531,3,0,01,00075AE1,00F2B9E0,15,2128DC34,3EEE0EFF,55,00000000,01CEA824,15,00F9000
0,4512440D,55,05B43821,31F1DC00,55,E445CB2B,B0124000,55,F5002D9C,20009540,55,71FC6B48
,79FFFFEF,D5,26DC0ACD,0C4875DA,55,F921EAAA,AAAAAAAA,AA*6F
$PAIR001,531,1*3D
$PAIR001,531,0*3C
$PAIR531,1*20

//BDS MEO
$PAIR531,3,0,06,E24057C7,A8001C72,F76059FC,8685FA88,107F813F,800745DF,C803A281,E24097
C7,484937E3,D485B7D5,F301EF12,430DE300,9D9F9027,2B9EA499,E240D7C7,61EEC134,2E8A87F
C,37FFD194,004AF287,85769D39,0EB28C70*3C
$PAIR001,531,1*3D
$PAIR001,531,0*3C
$PAIR531,1*20

//QZSS
$PAIR531,4,0,02,36A103,800000,000000,000000,000003,D15FCD,00FFFE,FFED74,D13542,1D1532,5
C75A1,1FD026,B477FA,2699CA,E91A23,5FCD7C,FCC900,029569,00321D,4E6276,CB7BBF,6D0E44
,FFDCD5,D1F2FC*6D
$PAIR001,531,1*3D
$PAIR001,531,0*3C
$PAIR531,1*20
```

#### NOTE

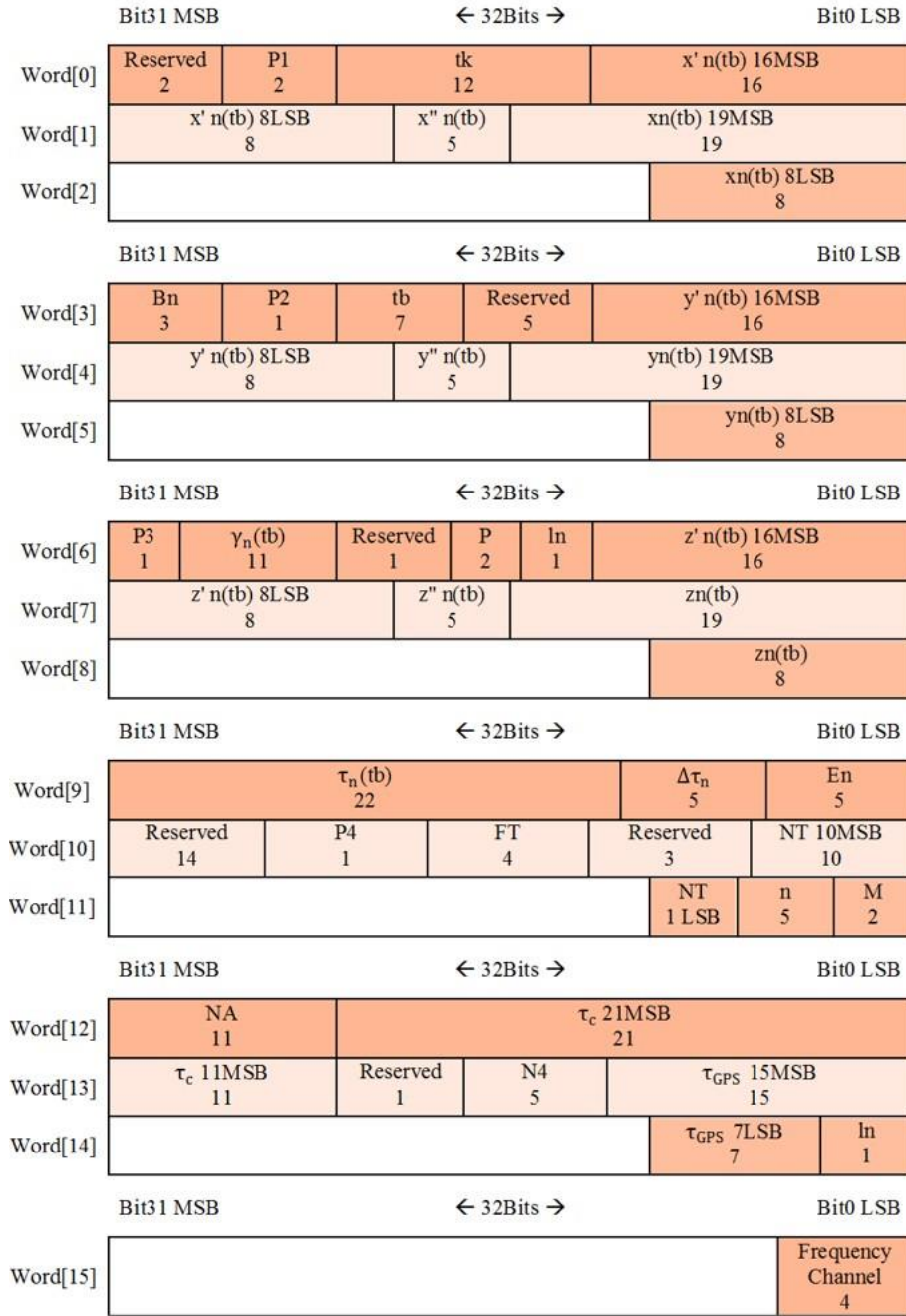
1. The satellite numbers of GLONASS satellites in NMEA 0183 start from 65. When sending the \$PAIR531 command, you need to align the satellite numbers in NMEA 0183 with the satellite

numbers of **\$PAIR531**, that is, subtract 64 from the GLONASS satellite number in NMEA 0183.

- The satellite numbers of QZSS satellites in NMEA 0183 start from 193. When sending the **\$PAIR531** command, you need to align the satellite numbers in NMEA 0183 with the satellite numbers of **\$PAIR531**, that is, subtract 192 from the QZSS satellite number in NMEA 0183.

|          | Bit23 MSB              | <-- 24 Bits --> |            | Bit0 LSB             |
|----------|------------------------|-----------------|------------|----------------------|
| Word[0]  | WN<br>10               | C/A<br>2        | URA<br>4   | HEALTH<br>6          |
| Word[1]  | L2P<br>1               | Reserved<br>23  |            |                      |
| Word[2]  | Reserved<br>24         |                 |            |                      |
| Word[3]  | Reserved<br>24         |                 |            |                      |
| Word[4]  | Reserved<br>16         |                 |            | TGD<br>8             |
| Word[5]  | IODC 8LSB<br>8         |                 | Toc<br>16  |                      |
| Word[6]  | Af2<br>8               |                 | Af1<br>16  |                      |
| Word[7]  | Af0<br>22              |                 |            | t<br>2               |
|          | Bit23 MSB              | <-- 24 Bits --> |            | Bit0 LSB             |
| Word[8]  | IODE<br>8              |                 | Crs<br>16  |                      |
| Word[9]  | $\Delta n$<br>16       |                 |            | M0 8MSB<br>8         |
| Word[10] | M0 24LSB<br>24         |                 |            |                      |
| Word[11] | Cuc<br>16              |                 |            | e 8MSB<br>8          |
| Word[12] | e 24LSB<br>24          |                 |            |                      |
| Word[13] | Cus<br>16              |                 |            | SQRT-A 8MSB<br>8     |
| Word[14] | SQRT-A 24LSB<br>24     |                 |            |                      |
| Word[15] | Toe<br>16              |                 | Fit<br>1   | AODO<br>5            |
|          |                        |                 |            | t<br>2               |
|          | Bit23 MSB              | <-- 24 Bits --> |            | Bit0 LSB             |
| Word[16] | Cic<br>16              |                 |            | $\Omega_0$ 8MSB<br>8 |
| Word[17] | $\Omega_0$ 24LSB<br>24 |                 |            |                      |
| Word[18] | Cis<br>16              |                 |            | i0 8MSB<br>8         |
| Word[19] | i0 24LSB<br>24         |                 |            |                      |
| Word[20] | Crc<br>16              |                 |            | w 8MSB<br>8          |
| Word[21] | w 24LSB<br>24          |                 |            |                      |
| Word[22] | $\Omega$<br>24         |                 |            |                      |
| Word[23] | IODE<br>8              |                 | IDOT<br>14 | t<br>2               |

**Figure 10: EPH Data Format – GPS and QZSS**


**Figure 11: EPH Data Format – GLONASS**

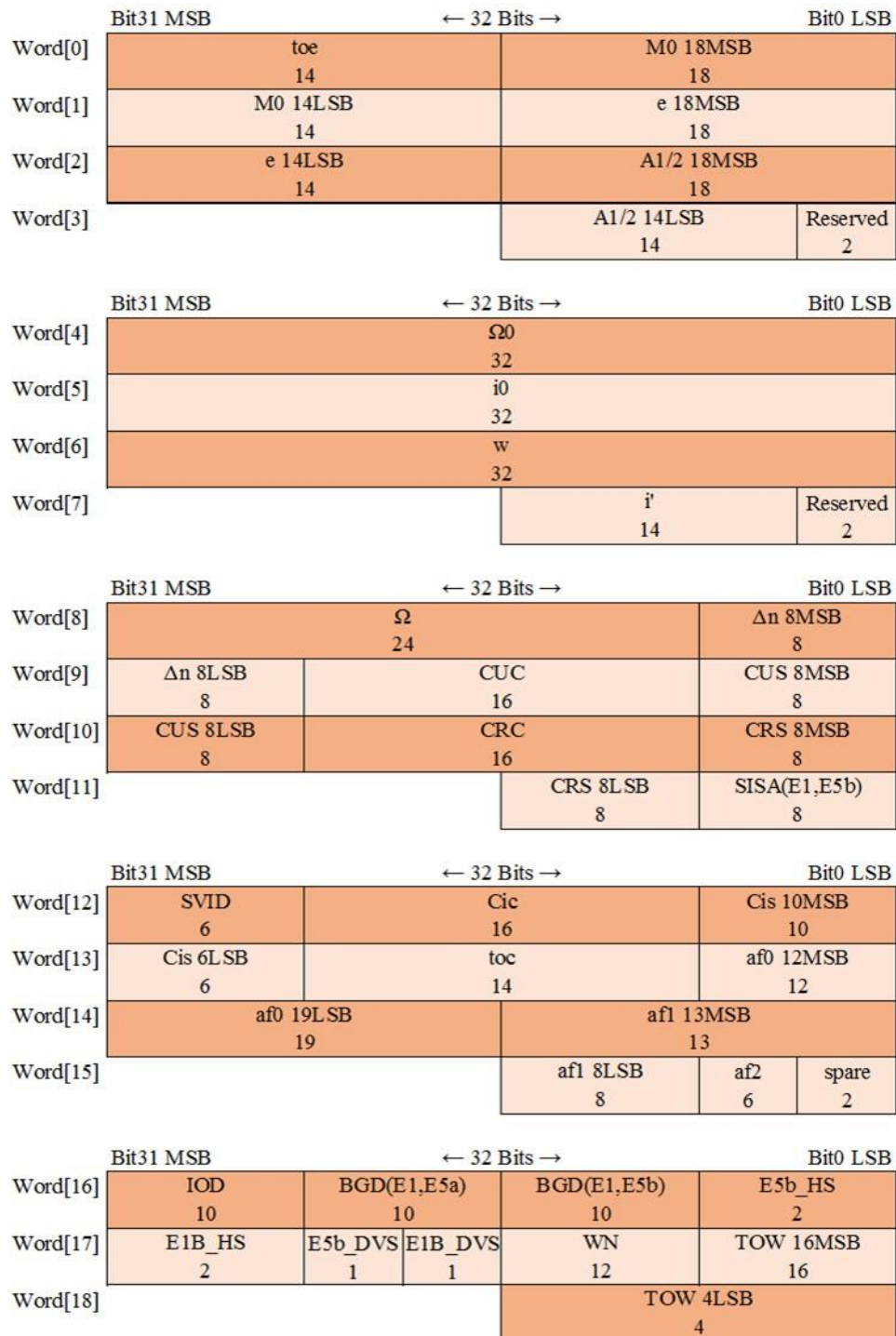


Figure 12: EPH Data Format – Galileo



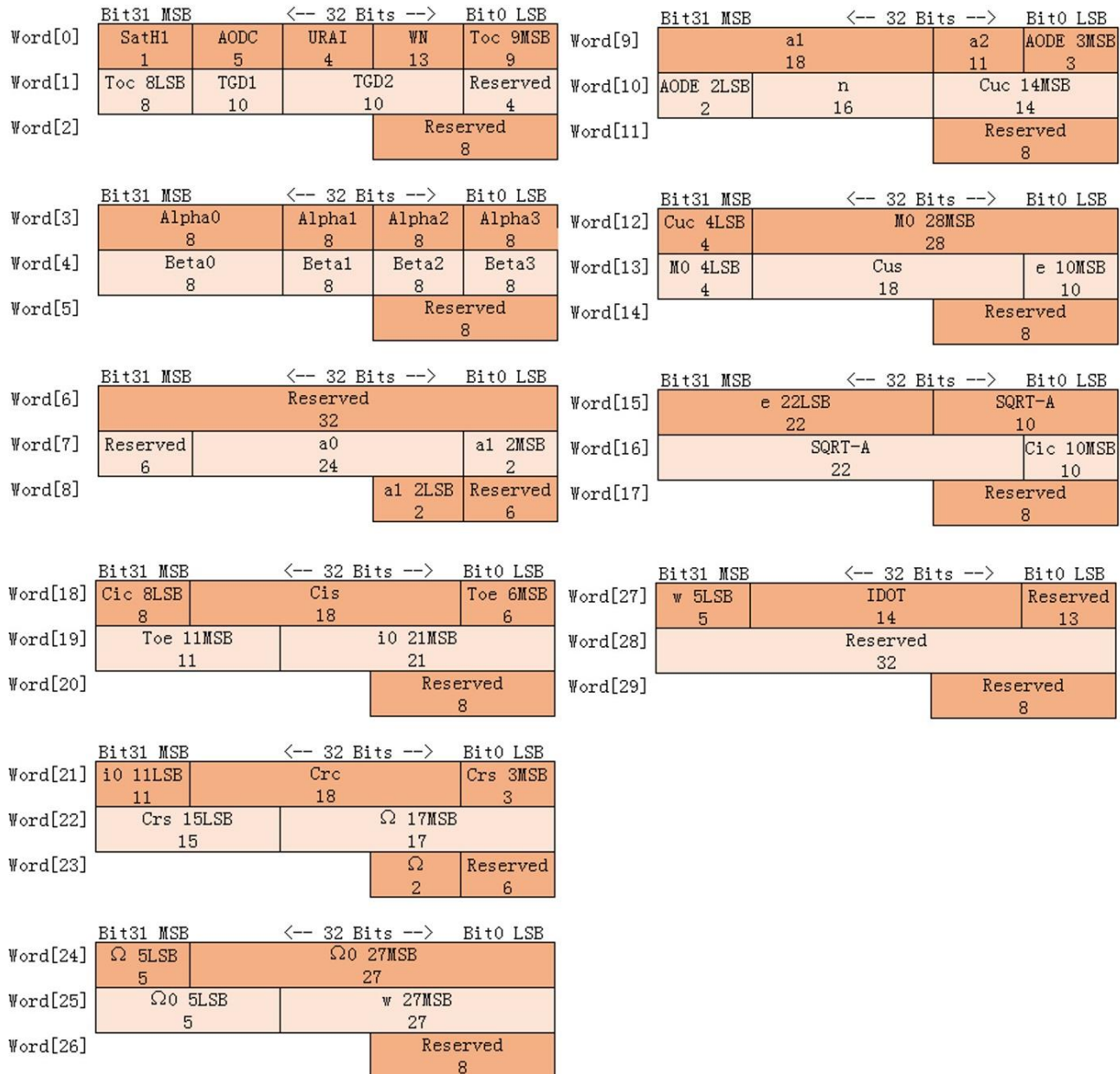


Figure 13: EPH Data Format – BDS GEO

|          | Bit31 MSB           | <-- 32 Bits --> |            |             |              | Bit0 LSB          |
|----------|---------------------|-----------------|------------|-------------|--------------|-------------------|
| Word[0]  | Pre 11              |                 | Reserved 4 | FraID 3     | SOW 14MSB 14 |                   |
| Word[1]  | SOW 6LSB 6          | SatH1 1         | AODC 5     | URAI 4      | WN 13        | toc 3MSB 3        |
| Word[2]  | toc 14LSB 14        |                 | TGD1 10    |             | TGD2 8MSB 8  |                   |
| Word[3]  | TGD2 2LSB 2         | Alpha0 8        | Alpha1 8   | Alpha2 8    |              | Alpha3 6MSB 6     |
| Word[4]  | Alpha3 2LSB 2       | Beta0 8         | Beta1 8    | Beta2 8     |              | Beta3 6MSB 6      |
| Word[5]  | Beta3 2LSB 2        | a2 11           |            | a0 19MSB 19 |              |                   |
| Word[6]  | a0 5LSB 5           | a1 22           |            |             |              | AODE 5            |
|          | Bit31 MSB           | <-- 32 Bits --> |            |             |              | Bit0 LSB          |
| Word[7]  | Pre 11              |                 | Reserved 4 | FraID 3     | SOW 14MSB 14 |                   |
| Word[8]  | SOW 6LSB 6          | Detan 16        |            |             |              | Cuc 10MSB 10      |
| Word[9]  | Cuc 8LSB 8          | M0 24MSB 24     |            |             |              |                   |
| Word[10] | M0 8LSB 8           | e 24MSB 24      |            |             |              |                   |
| Word[11] | e 8LSB 8            | Cus 18          |            |             |              | Crc 6MSB 6        |
| Word[12] | Crc 12LSB 12        |                 | Crs 18     |             |              | SQRT-A 2MSB 2     |
| Word[13] | SQRT-A 30LSB 30     |                 |            |             |              | Toe 2MSB 2        |
|          | Bit31 MSB           | <-- 32 Bits --> |            |             |              | Bit0 LSB          |
| Word[14] | Pre 11              |                 | Reserved 4 | FraID 3     | SOW 14MSB 14 |                   |
| Word[15] | SOW 6LSB 6          | Toe 15LSB 15    |            |             |              | i0 11MSB 11       |
| Word[16] | i0 21LSB 21         |                 |            |             |              | Cic 11MSB 11      |
| Word[17] | Cic 7LSB 7          | $\Omega$ 24     |            |             |              | Cis 1MSB 1        |
| Word[18] | Cis 17LSB 17        |                 | IDOT 14    |             |              | $\Omega_0$ 1MSB 1 |
| Word[19] | $\Omega_0$ 31LSB 31 |                 |            |             |              | $\omega$ 1MSB 1   |
| Word[20] | $\omega$ 31LSB 31   |                 |            |             |              | Reserved 1        |

Figure 14: EPH Data Format – BDS MEO

### 3.1.3. PAIR532: PAIR\_EPH\_GET\_DATA

Gets the EPH data for a specified satellite.

**Type:**

Command

### Synopsis:

```
$PAIR532,<Constellation>,<Signal_ID>,<SV_ID>*<Checksum><CR><LF>
```

### Parameter:

| Field           | Format  | Unit | Description                                                                                                                                                    |
|-----------------|---------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Constellation> | Numeric | -    | GNSS System ID.<br>0 = GPS<br>1 = GLONASS<br>2 = Galileo<br>3 = BDS<br>4 = QZSS                                                                                |
| <Signal_ID>     | Numeric | -    | Signal ID. Always "0".<br>0 = L1                                                                                                                               |
| <SV_ID>         | Numeric | -    | SV ID. Ranges for different constellations are as below.<br>GPS: 1–32<br>GLONASS: 1–24<br>Galileo: 1–36<br>BDS: 1–63 (GEO: 1–5, 59–63; MEO: 6–58)<br>QZSS: 1–7 |

### Result:

Returns **\$PAIR001** message and the query result.

### Query result message format:

```
$PAIR532,<Constellation>,<Signal_ID>,<SV_ID>,<EPH_data>*<Checksum><CR><LF>
```

### Parameters included in the result:

| Field           | Format  | Unit | Description                                                                                             |
|-----------------|---------|------|---------------------------------------------------------------------------------------------------------|
| <Constellation> | Numeric | -    | GNSS System ID.<br>0 = GPS<br>1 = GLONASS<br>2 = Galileo<br>3 = BDS<br>4 = QZSS                         |
| <Signal_ID>     | Numeric | -    | Signal ID. Always "0".<br>0 = L1                                                                        |
| <SV_ID>         | Numeric | -    | SV ID. Ranges for different constellations are as below.<br>GPS: 1–32<br>GLONASS: 1–24<br>Galileo: 1–36 |

| Field      | Format      | Unit | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------|-------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <EPH_Data> | Hexadecimal | -    | BDS: 1–63 (GEO: 1–5, 59–63; MEO: 6–58)<br>QZSS: 1–7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|            |             |      | EPH data. The format of EPH data for different constellations is as follows.<br><b>GPS and QZSS:</b> Word[0]–Word[23]. The 24-word EPH data cover words 3–10 from subframes 1–3 in the GPS or QZSS Navigation Message.<br>The raw 30-bit data of each word have been logically shifted 6 bits to the right to remove the 6 parity bits leaving the 24-bit data. For details on data format, see <a href="#">Figure 10: EPH Data Format – GPS and QZSS</a> .                                                                                                                               |
|            |             |      | <b>GLONASS:</b> Word[0]–Word[15]. The first 15-word EPH data cover strings 1–5 in the GLONASS Navigation Message. The Last word, Word[15], indicates the frequency channel (range: 1–14).<br>For details on data format, see <a href="#">Figure 11: EPH Data Format – GLONASS</a> .                                                                                                                                                                                                                                                                                                       |
|            |             |      | <b>Galileo:</b> Word[0]–Word[18]. The 19-word EPH data cover 16 words from the subframes 1–4 in the Galileo Navigation Message and 3 words from subframe 5 with BGD, signal health status, data validity, and GST.<br>Note that only Word[0]–Word[15] need to be input when sending ephemeris, while Word[0]–Word[18] will be provided when getting ephemeris.<br>The Page Type (6-bit) and IOD <sub>nav</sub> (10-bit) have been removed and the rest bits of each word are shifted to the right. For details on data format, see <a href="#">Figure 12: EPH Data Format – Galileo</a> . |
|            |             |      | <b>BDS GEO:</b> Word[0]–Word[29]. The 30-word EPH data cover pages 1–10 from subframe 1 in the BDS GEO D2 Navigation Message.<br>The raw 30-bit data of each word have been logically shifted 8 bits to the right to remove the 8 parity bits. For details on data format, see <a href="#">Figure 13: EPH Data Format – BDS GEO</a> .                                                                                                                                                                                                                                                     |
|            |             |      | <b>BDS MEO:</b> Word[0]–Word[20]. The 21-word EPH data cover subframes 1–3 in the BDS MEO D1 Navigation Message.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Field | Format | Unit | Description                                                                                                                                                                                          |
|-------|--------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |        |      | The raw 30-bit data of each word have been logically shifted 8 bits to the right to remove the 8 parity bits. For details on data format, see <a href="#">Figure 14: EPH Data Format – BDS MEO</a> . |

#### Example:

```
//GPS
$PAIR532,0,0,8*2A
$PAIR001,532,1*3E
$PAIR001,532,0*3F
$PAIR532,0,0,08,3E5000,72962C,B66E4F,B1894A,86D40B,2B7080,00FFF0,EAA8AB,2B0771,33CE2
7,3ABE45,06A904,B7489D,0748A1,0CA1E6,70807F,FFE387,4CC730,004B26,EDAA61,267D0C,5270
5A,FFA33F,2BFD9F*61

//GLONASS
$PAIR532,1,0,10*12
$PAIR001,532,1*3E
$PAIR001,532,0*3F
$PAIR532,1,0,10,0521129B,92166C00,000000E4,15208EF0,311172BE,0000001F,C046B222,180D543
C,000000E4,08656C00,0002A01C,000000A9,07300000,01A40000,00000020,00000001*16

//Galileo
$PAIR532,2,0,10*11
$PAIR001,532,1*3E
$PAIR001,532,0*3F
$PAIR532,2,0,10,4B29379F,7BD80009,90C2A813,000089D3,A9823149,28BD1CC8,D9B6AF27,00000
619,FFC67C1A,1901A414,D2124102,0000696B,280013FF,FD2CAFEC,650E3FFF,00003301,1840100
C,04FB46E9,00000007*4C

//BDS GEO
$PAIR532,3,0,3*22
$PAIR001,532,1*3E
$PAIR001,532,0*3F
$PAIR532,3,0,03,00075AE1,0005FAF0,15,2128DC34,3EEE0EFF,55,00000000,03CF3F80,15,1C02400
0,446B81F4,55,2E86E526,B2045801,55,E57F9B2B,AD088801,55,3000501C,20009517,55,B9DC4E08
,36CDFFF1,55,2ED49A12,2F25A36E,55,111D6AAA,AAAAAAAA,AA*18

//BDS MEO
$PAIR532,3,0,16*16
$PAIR001,532,1*3E
$PAIR001,532,0*3F
```

```
$PAIR532,3,0,16,E24051BA,E8001D7A,3283E50C,4B7CBF81,8D807DC4,0006B895,A7F3EBC1,E240
91BA,8822EFE8,28E3BB4E,46037EDE,EE2BD0B9,CC6FB32F,2B9E8FAD,E240D1BA,A0650139,BB
2CBFFF,0FFFD986,0034FEEC,3E4FAE87,46D72F64*42
```

```
//QZSS
```

```
$PAIR532,4,0,2*24
```

```
$PAIR001,532,1*3E
```

```
$PAIR001,532,0*3F
```

```
$PAIR532,4,0,02,3EE107,800000,000000,000000,000003,654812,00FFFD,FFCABC,651777,07C1B6,4
401CD,0DFE26,6AF306,F6C2CA,EB5349,48127E,03295B,CAB930,F78F1C,FDCB97,269EBF,D7368F
,FFFA98,651E36*65
```

### 3.1.4. PAIR533: PAIR\_EPH\_CLEAR

Clears the EPH data in the critical memory area.

#### Type:

Command

#### Synopsis:

```
$PAIR533*<Checksum><CR><LF>
```

#### Parameter:

None

#### Result:

Returns **\$PAIR001** message.

#### Example:

```
$PAIR533*3F
```

```
$PAIR001,533,0*3E
```

### 3.1.5. PAIR534: PAIR\_EPH\_NOTIFY\_ENABLE

Enables/disables the notification that updated EPH data is available.

#### Type:

Command

### Synopsis:

```
$PAIR534,<Enabled>*<Checksum><CR><LF>
```

### Parameter:

| Field     | Format  | Unit | Description                                                                                      |
|-----------|---------|------|--------------------------------------------------------------------------------------------------|
| <Enabled> | Numeric | -    | Enable/Disable the notification that updated EPH data is available.<br>0 = Disable<br>1 = Enable |

### Result:

Returns **\$PAIR001** message.

### Example:

```
$PAIR534,1*25
$PAIR001,534,0*39
```

## 3.1.6. PAIR535: PAIR\_EPH\_NOTIFY

Outputs the notification that updated EPH data is available.

### Type:

Output

### Synopsis:

```
$PAIR535,<Constellation>,<Signal_ID>,<SV_Information>*<Checksum><CR><LF>
```

### Parameter:

| Field            | Format      | Unit | Description                                                                                      |
|------------------|-------------|------|--------------------------------------------------------------------------------------------------|
| <Constellation>  | Numeric     | -    | GNSS System ID.<br>0 = GPS<br>1 = GLONASS<br>2 = Galileo<br>3 = BDS<br>4 = QZSS                  |
| <Signal_ID>      | Numeric     | -    | Signal ID. Always "0".<br>0 = L1                                                                 |
| <SV_Information> | Hexadecimal | -    | SV ID bitmap to indicate the satellite(s) whose updated EPH data are available. Length: 64 bits. |

### Example:

```
$PAIR535,0,0,0000000010080000*1C
```

//The binary format of 0000000010080000 is: 0001 0000 0000 1000 0000 0000 0000 0000. The corresponding satellite numbers are 20 and 29.

## 3.2. ALM Related Messages

### 3.2.1. PAIR550: PAIR\_ALM\_GET\_STATUS

Queries satellites whose valid ALM data will be available for a specified future time period.

#### Type:

Command

#### Synopsis:

```
$PAIR550,<Constellation>,<Time_Interval>*<Checksum><CR><LF>
```

#### Parameter:

| Field           | Format  | Unit | Description                                                                     |
|-----------------|---------|------|---------------------------------------------------------------------------------|
| <Constellation> | Numeric | -    | GNSS System ID.<br>0 = GPS<br>1 = GLONASS<br>2 = Galileo<br>3 = BDS<br>4 = QZSS |
| <Time_Interval> | Numeric | Day  | Specified future time period. Range: 1–91.                                      |

#### Result:

Returns **\$PAIR001** message and the query result.

#### Query result message format:

```
$PAIR550,<Constellation>,<L1_SV>*<Checksum><CR><LF>
```

#### Parameters included in the result:

| Field           | Format  | Unit | Description     |
|-----------------|---------|------|-----------------|
| <Constellation> | Numeric | -    | GNSS System ID. |



| Field   | Format      | Unit | Description                                                                                                                                                                                                                                                                                                                                                         |
|---------|-------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |             |      | 0 = GPS<br>1 = GLONASS<br>2 = Galileo<br>3 = BDS<br>4 = QZSS                                                                                                                                                                                                                                                                                                        |
| <L1_SV> | Hexadecimal | -    | Valid ALM SV ID. Field Lengths for different constellations are as follows.<br>GPS: 8 bytes<br>GLONASS: 6 bytes<br>Galileo: 16 bytes<br>BDS: 16 bytes<br>QZSS: 8 bytes<br>Note that each bit represents a satellite, and SV ID counting starts from the least significant bit. Take GPS as an example. If Bit 0 is 1, the ALM data of GPS satellite 1 is available. |

#### Example:

```
$PAIR550,0,10*0B
```

```
$PAIR001,550,1*3A
```

```
$PAIR001,550,0*3B
```

```
$PAIR550,0,FFFFFFFE*09
```

//The binary format of FFFFFFFE is: 1111 1111 1111 1111 1111 1111 1111 1110. The corresponding satellite numbers are 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, and 32.

### 3.2.2. PAIR551: PAIR\_ALM\_SET\_DATA

Sends ALM data to GNSS module.

#### Type:

Input

#### Synopsis:

```
$PAIR551,<Constellation>,<Signal_ID>,<SV_ID>,<WeekNo>,<ALM_Data>*<Checksum><CR><LF>
```

## Parameter:

| Field           | Format      | Unit | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------|-------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Constellation> | Numeric     | -    | GNSS System ID.<br>0 = GPS<br>1 = GLONASS<br>2 = Galileo<br>3 = BDS<br>4 = QZSS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <Signal_ID>     | Numeric     | -    | Signal ID. Always "0".<br>0 = L1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <SV_ID>         | Numeric     | -    | SV ID. Ranges for different constellations are as below.<br>GPS: 1–32<br>GLONASS: 1–24<br>Galileo: 1–36<br>BDS: 1–63 (GEO: 1–5, 59–63; MEO: 6–58)<br>QZSS: 1–7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <WeekNo>        | Hexadecimal | -    | Almanac reference week number.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <ALM_Data>      | Hexadecimal | -    | <p>ALM data. The format of ALM data for different constellations is as follows.</p> <p><b>GPS and QZSS:</b> Word[0]–Word[7]. The 8-word ALM data cover pages 1–24 of subframe 5, as well as pages 2–5 and 7–9 of subframe 4 in the GPS or QZSS Navigation Message.</p> <p>The raw 30-bit data of each word have been logically shifted 6 bits to the right to remove the 6 parity bits leaving the 24 data bits. For details on data format, see <a href="#">Figure 15: ALM Data Format – GPS and QZSS</a>.</p> <p><b>GLONASS:</b> Word[0]–Word[5]. The 6-word ALM data cover strings 6–15 of the GLONASS Navigation Message. For details on data format, see <a href="#">Figure 16: ALM Data Format – GLONASS</a>.</p> <p><b>Galileo:</b> Word[0]–Word[3]. The 4-word ALM data cover subframes 7–10 in the Galileo Navigation Message. For details on data format, see <a href="#">Figure 17: ALM Data Format – Galileo</a>.</p> <p><b>BDS:</b> Word[0]–Word[6]. The 7-word ALM data cover pages 37–60 and 95–100 in subframe 5 (for GEO), pages 1–24 in subframe 4, and pages 1–6 in subframe</p> |

| Field | Format | Unit | Description                                                                                                                              |
|-------|--------|------|------------------------------------------------------------------------------------------------------------------------------------------|
|       |        |      | 5 (for MEO).<br>The 11 parity bits have been removed. For details on data format, see <a href="#">Figure 18: ALM Data Format – BDS</a> . |

**Result:**

Returns **\$PAIR001** message and the query result.

**Query result message format:**

```
$PAIR551,<Status>*<Checksum><CR><LF>
```

**Parameter included in the result:**

| Field    | Format  | Unit | Description                  |
|----------|---------|------|------------------------------|
| <Status> | Numeric | -    | 0 = Failed<br>1 = Successful |

**Example:**

```
//GPS
$PAIR551,0,0,02,08F9,4285A8,90104B,FD3F00,A10EC7,AFD145,CCB416,3569A4,BE0058*30
$PAIR001,551,1*3B
$PAIR001,551,0*3A
$PAIR551,1*26

//GLONASS
$PAIR551,1,0,02,0012,16C908E0,0169A56C,000000A2,97EDF638,42502DE6,000000DD*35
$PAIR001,551,1*3B
$PAIR001,551,0*3A
$PAIR551,1*26

//Galileo
$PAIR551,2,0,02,04FB,0080221A,5FF94A91,BFC06E34,00D00020*47
$PAIR001,551,1*3B
$PAIR001,551,0*3A
$PAIR551,1*26

//BDS
$PAIR551,3,0,01,03AD,E2411C20,48072BAF,E40039DE,32B1F01C,48911188,00E60244,D55D5F97*6
9
$PAIR001,551,1*3B
$PAIR001,551,0*3A
```

```
$PAIR551,1*26
```

```
//QZSS
```

```
$PAIR551,4,0,02,08FA,C27AD8,15D037,FF3B01,CAEA53,60CCA9,BFC206,43CC9F,FF001C*4A
```

```
$PAIR001,551,1*3B
```

```
$PAIR001,551,0*3A
```

```
$PAIR551,1*26
```

### NOTE

1. The satellite numbers of GLONASS satellites in NMEA 0183 start from 65. When sending the **\$PAIR551** command, you need to align the satellite numbers in NMEA 0183 with the satellite numbers of **\$PAIR551**, that is, subtract 64 from the GLONASS satellite number in NMEA 0183.
2. The satellite numbers of QZSS satellites in NMEA 0183 start from 193. When sending the **\$PAIR551** command, you need to align the satellite numbers in NMEA 0183 with the satellite numbers of **\$PAIR551**, that is, subtract 192 from the QZSS satellite number in NMEA 0183.

|         | Bit23 MSB                 | ← 24Bits →            |                  | Bit0 LSB                            |
|---------|---------------------------|-----------------------|------------------|-------------------------------------|
| Word[0] | Data ID<br>2              | SV ID<br>6            | e<br>16          |                                     |
| Word[1] | Toa<br>8                  |                       | $\delta_i$<br>16 |                                     |
| Word[2] | $\dot{\Omega}$<br>16      |                       |                  | SV Health<br>8                      |
| Word[3] | $\sqrt{A}$<br>24          |                       |                  |                                     |
| Word[4] | $\Omega_0$<br>24          |                       |                  |                                     |
| Word[5] | $\omega$<br>24            |                       |                  |                                     |
| Word[6] | $M_0$<br>24               |                       |                  |                                     |
| Word[7] | af <sub>0</sub> 8MSB<br>8 | af <sub>1</sub><br>11 |                  | af <sub>0</sub> 3LSB<br>3<br>t<br>2 |

Figure 15: ALM Data Format – GPS and QZSS

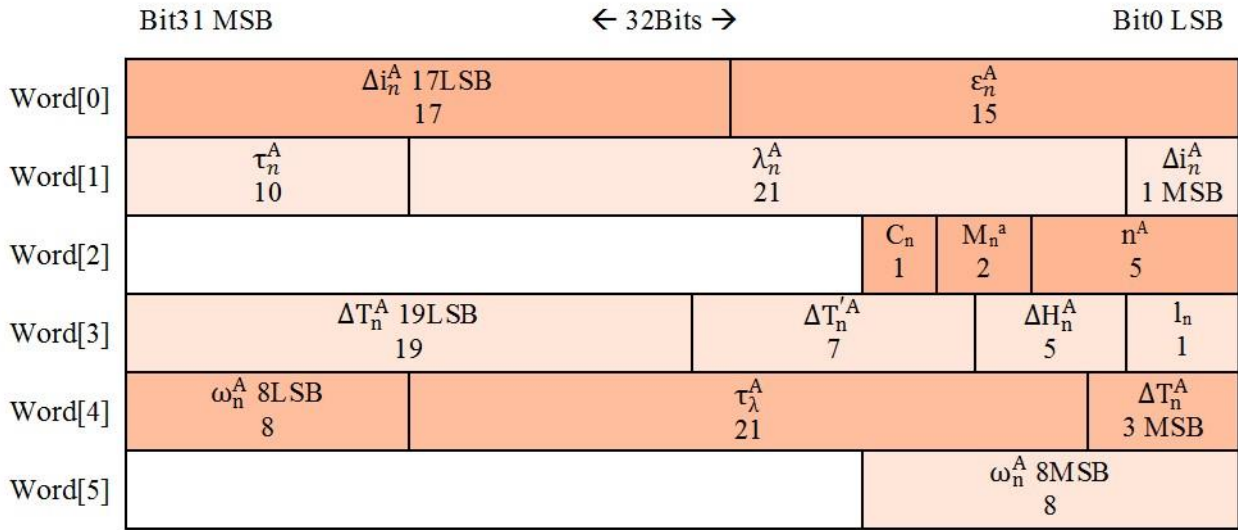


Figure 16: ALM Data Format – GLONASS

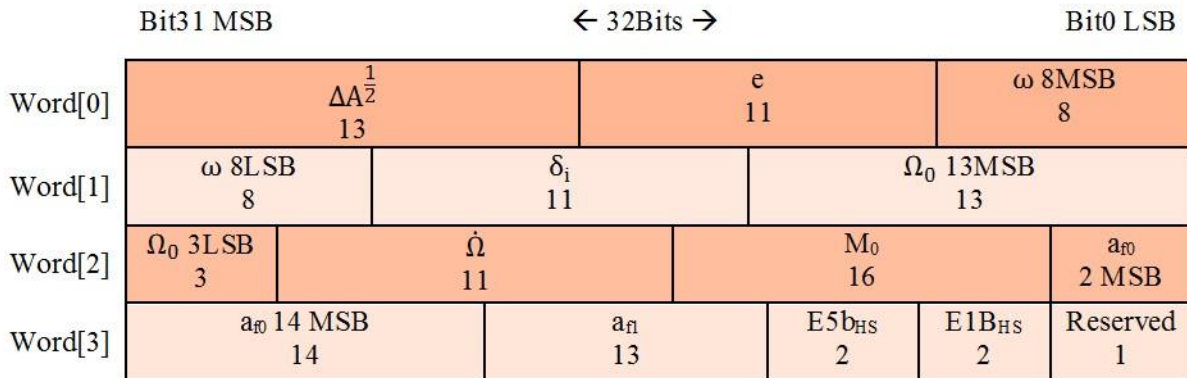
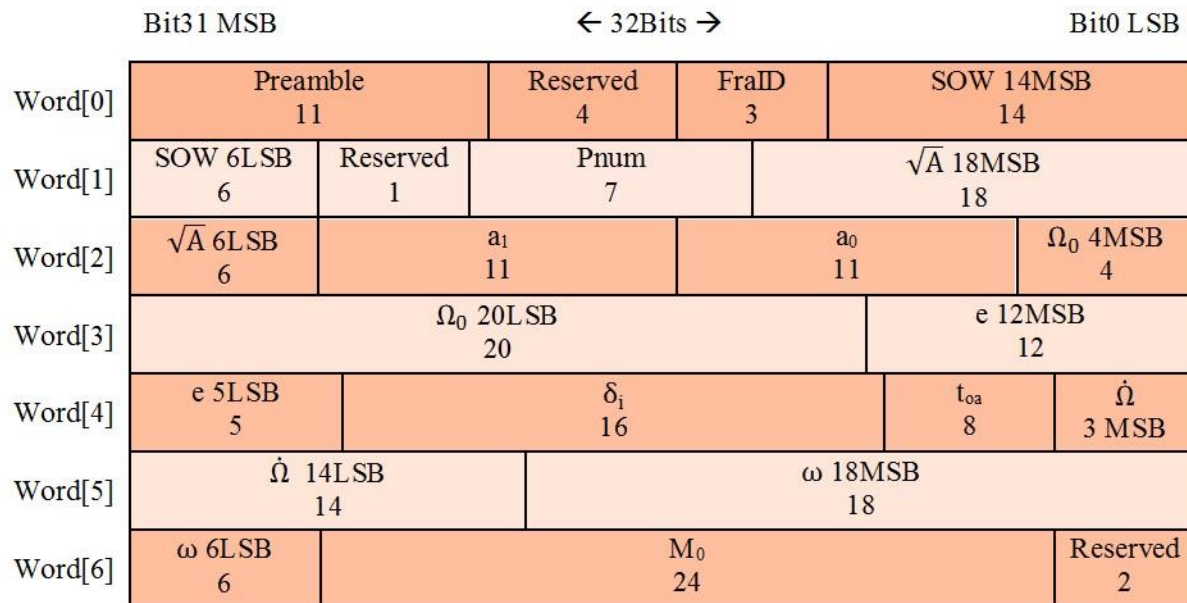


Figure 17: ALM Data Format – Galileo



**Figure 18: ALM Data Format – BDS**

### 3.2.3. PAIR552: PAIR\_ALM\_GET\_DATA

Gets the ALM data of the specified satellite.

#### Type:

Command

#### Synopsis:

```
$PAIR552,<Constellation>,<Signal_ID>,<SV_ID>*<Checksum><CR><LF>
```

#### Parameter:

| Field           | Format  | Unit | Description                                                                     |
|-----------------|---------|------|---------------------------------------------------------------------------------|
| <Constellation> | Numeric | -    | GNSS System ID.<br>0 = GPS<br>1 = GLONASS<br>2 = Galileo<br>3 = BDS<br>4 = QZSS |
| <Signal_ID>     | Numeric | -    | Signal ID. Always "0".<br>0 = L1                                                |
| <SV_ID>         | Numeric | -    | SV ID. Ranges for different constellations are as below.<br>GPS: 1–32           |

| Field | Format | Unit | Description                                                                           |
|-------|--------|------|---------------------------------------------------------------------------------------|
|       |        |      | GLONASS: 1–24<br>Galileo: 1–36<br>BDS: 1–63 (GEO: 1–5, 59–63; MEO: 6–58)<br>QZSS: 1–7 |

#### Result:

Returns **\$PAIR001** message and the query result.

#### Query result message format:

```
$PAIR552,<Constellation>,<Signal_ID>,<SV_ID>,<WN>,<ALM_Data>*<Checksum><CR><LF>
```

#### Parameters included in the result:

| Field           | Format      | Unit | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------|-------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Constellation> | Numeric     | -    | GNSS System ID.<br>0 = GPS<br>1 = GLONASS<br>2 = Galileo<br>3 = BDS<br>4 = QZSS                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <Signal_ID>     | Numeric     | -    | Signal ID. Always “0”.<br>0 = L1                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <SV_ID>         | Numeric     | -    | SV ID. Ranges for different constellations are as below.<br>GPS: 1–32<br>GLONASS: 1–24<br>Galileo: 1–36<br>BDS: 1–63 (GEO: 1–5, 59–63; MEO: 6–58)<br>QZSS: 1–7                                                                                                                                                                                                                                                                                                                                    |
| <WN>            | Hexadecimal | Week | Almanac reference week number.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <ALM_Data>      | Hexadecimal | -    | ALM data. The format of ALM data for different constellations is as follows.<br><b>GPS and QZSS:</b> Word[0]–Word[7]. The 8-word ALM data cover pages 1–24 of subframe 5, as well as pages 2–5 and 7–9 of subframe 4 in the GPS or QZSS Navigation Message.<br>The raw 30-bit data of each word have been logically shifted 6 bits to the right to remove the 6 parity bits leaving the 24 data bits. For details on data format, see <a href="#">Figure 15: ALM Data Format – GPS and QZSS</a> . |

| Field | Format | Unit | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------|--------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |        |      | <p><b>GLONASS:</b> Word[0]–Word[5]. The 6-word ALM data cover strings 6–15 of the GLONASS Navigation Message. For details on data format, see <a href="#">Figure 16: ALM Data Format – GLONASS</a>.</p> <p><b>Galileo:</b> Word[0]–Word[3]. The 4-word ALM data cover subframes 7–10 in the Galileo Navigation Message. For details on data format, see <a href="#">Figure 17: ALM Data Format – Galileo</a>.</p> <p><b>BDS:</b> Word[0]–Word[6]. The 7-word ALM data cover pages 37–60 and 95–100 in subframe 5 (for GEO), pages 1–24 in subframe 4, and pages 1–6 in subframe 5 (for MEO).<br/>The 11 parity bits have been removed. For details on data format, see <a href="#">Figure 18: ALM Data Format – BDS</a>.</p> |

#### Example:

```
//GPS
$PAIR552,0,0,8*2C
$PAIR001,552,1*38
$PAIR001,552,0*39
$PAIR552,0,0,08,08FA,484B97,0F0876,FD4C00,A10C42,82533D,0C60FC,EC2109,EA0014*42

//GLONASS
$PAIR552,1,0,16*12
$PAIR001,552,1*38
$PAIR001,552,0*39
$PAIR555,1,0,16,0012,13A98271,81948442,000000B0,9802F37E,BA2092F6,000000FB*4A

//Galileo
$PAIR552,2,0,10*17
$PAIR001,552,1*38
$PAIR001,552,0*39
$PAIR552,2,0,10,04FB,006801DB,700EB530,5FC9613B,FB1BFFE0*49

//BDS
$PAIR552,3,0,16*10
$PAIR001,552,1*38
$PAIR001,552,0*39
$PAIR552,3,0,16,03AD,E2415C22,64D32B99,BBFD6D62,90F8F1BB,7065A18F,FD7E8C62,BF57867D
*1D
```



```
//QZSS
$PAIR552,4,0,2*22
$PAIR001,552,1*38
$PAIR001,552,0*39
$PAIR552,4,0,02,08FB,C27ADC,7FCFD6,FF1E01,CAEAD7,5BC3FA,BFDD50,52BA0D,FF0019*4D
```

### 3.2.4. PAIR553: PAIR\_ALM\_CLEAR

Clears the ALM data in the critical memory area.

#### Type:

Command

#### Synopsis:

```
$PAIR553*<Checksum><CR><LF>
```

#### Parameter:

None

#### Result:

Returns **\$PAIR001** message.

#### Example:

```
$PAIR553*39
$PAIR001,553,0*38
```

# 4 Appendix A References

**Table 10: Related Documents**

| Document Name                                                                                        |
|------------------------------------------------------------------------------------------------------|
| [1] <a href="#">Quectel LC26G&amp;LC26G-T&amp;LC76G&amp;LC86G Series GNSS Protocol Specification</a> |
| [2] <a href="#">Quectel LC26G(AB) Hardware Design</a>                                                |
| [3] <a href="#">Quectel LC76G Series Hardware Design</a>                                             |
| [4] <a href="#">Quectel LC86G Series Hardware Design</a>                                             |
| [5] <a href="#">Quectel LC26G&amp;LC76G&amp;LC86G Series AGNSS Application Note</a>                  |

**Table 11: Terms and Abbreviations**

| Abbreviation | Description                        |
|--------------|------------------------------------|
| ALP          | Adaptive Low Power                 |
| ALM          | Almanac                            |
| BDS          | BeiDou Navigation Satellite System |
| BGD          | Broadcast Group Delay              |
| CPU          | Central Processing Unit            |
| DSP          | Digital Signal Processing          |
| EPH          | Ephemeris                          |
| EPO          | Extended Prediction Orbit          |
| FLP          | Fitness Low Power                  |
| GEO          | Geostationary Orbit                |
| GLP          | GPS Low Power                      |

| Abbreviation | Description                                |
|--------------|--------------------------------------------|
| GNSS         | Global Navigation Satellite System         |
| GPIO         | General-Purpose Input/Output               |
| GPS          | Global Positioning System                  |
| GST          | Galileo System Time                        |
| ID           | Identifier                                 |
| IOD          | Issue of Data                              |
| LSB          | Least Significant Bit                      |
| ME           | Measurement Engine                         |
| MEO          | Medium Earth Orbit                         |
| MSB          | Most Significant Bit                       |
| PE           | Positioning Engine                         |
| RF           | Radio Frequency                            |
| SRAM         | Static Random Access Memory                |
| SV           | Satellites in View                         |
| TCXO         | Temperature Compensated Crystal Oscillator |
| TTF          | Time to First Fix                          |

# 5 Appendix B Special Characters

Table 12: Special Characters

| Special Character | Definition                                                                 |
|-------------------|----------------------------------------------------------------------------|
| <CR>              | Carriage return character.                                                 |
| <LF>              | Line feed character.                                                       |
| <...>             | Parameter name. Angle brackets do not appear in the message.               |
| [...]             | Optional field of a message. Square brackets do not appear in the message. |
| {...}             | Repeated field of a message. Curly brackets do not appear in the message.  |
| <u>Underline</u>  | Default setting of a parameter.                                            |